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Delayed Hydrogen Exsolution in Sub-Neptune Exoplanets under Magma-Envelope Coupling

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of F. Laga & S. Vanhulle (2025)

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All data, code and figures are reproducible from the directory `research/spinodal_onset` of the Atmodeller fork on the branch `feat/spinodal-metastability-study`. Source data of Gilmore & Stixrude (2026) are used under the Creative Commons Attribution 4.0 licence.

Chapters 1 and 2 reuse wording from the bachelor thesis of F. Laga and S. Vanhulle, *Atmospheric Composition of Sub-Neptune Exoplanets under Magma-Envelope Coupling* (KU Leuven, 2025).

Preface

We would like to thank Dr. Kaustubh Hakim for his guidance and teachings during the bachelor thesis on which this work builds. We would also like to thank Siemen Vanhulle for the joint thesis work, from which parts of the text are reused.

We acknowledge the use of AI tools for writing code, running the analysis and drafting this text. Every result, interpretation and conclusion must be checked by the authors before submission.

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Abstract

Sub-Neptune exoplanets represent the most abundant class of planets detected in the known exoplanet population. Despite their ubiquity, their fundamental properties remain poorly characterized. In particular, first-principles simulations suggest that hydrogen and silicate are completely miscible at the magma-atmosphere boundary of many young sub-Neptunes, so that hydrogen exsolves as the planet cools, but the timing of this exsolution remains largely unexplored.

In this work, we investigate the thermodynamic limits on hydrogen exsolution from H_2–$MgSiO_3$ mixtures using Atmodeller, a Python-based framework designed for coupled atmosphere-interior modeling. We implement the final 2026 mixing model with a composition-consistent basis, compare the equilibrium onset of exsolution (the binodal) with the loss of local stability (the spinodal), and recalibrate four smooth model classes against the source simulations.

With the published fit, a parcel with 3 wt% hydrogen at 2 GPa enters the miscibility gap at 3356 K but becomes unstable only at 2942 K; if nucleation is suppressed, up to 65.5% of its hydrogen is withheld over this 414 K interval. This delay is not constrained by the source data: refits of the same data give median intervals of 42 to 220 K. Equilibrium miscibility data therefore do not determine when hydrogen separates. A direct measurement of the composition curvature would fix the spinodal to within about 25 K.

Samenvatting

Sub-Neptunus-exoplaneten vormen de meest talrijke klasse binnen de huidige exoplaneetpopulatie. Ondanks hun alomtegenwoordigheid blijven hun fundamentele eigenschappen echter slecht gekarakteriseerd. Simulaties vanuit eerste principes suggereren dat waterstof en silicaat volledig mengbaar zijn aan de grens tussen magma en atmosfeer van veel jonge sub-Neptunussen. Waterstof ontmengt dan wanneer de planeet afkoelt, maar het tijdstip van deze ontmenging is tot op heden nauwelijks onderzocht.

In dit werk bestuderen wij de thermodynamische grenzen van de ontmenging van waterstof uit H_2–$MgSiO_3$-mengsels met behulp van Atmodeller, een in Python ontwikkeld raamwerk voor gekoppelde atmosfeer-mantelmodellering. Wij implementeren het definitieve mengmodel uit 2026 met een consistente samenstellingsbasis, vergelijken het begin van de ontmenging in evenwicht (de binodaal) met het verlies van lokale stabiliteit (de spinodaal), en kalibreren vier gladde modelklassen opnieuw op de oorspronkelijke simulaties.

Met de gepubliceerde fit komt een mengsel met 3 massa% waterstof bij 2 GPa de mengbaarheidskloof binnen bij 3356 K, maar wordt het pas instabiel bij 2942 K; als kiemvorming onderdrukt wordt, blijft tot 65.5% van de waterstof achter over dit interval van 414 K. De oorspronkelijke gegevens leggen deze vertraging niet vast: nieuwe fits aan dezelfde gegevens geven mediane intervallen van 42 tot 220 K. Evenwichtsgegevens over mengbaarheid bepalen dus niet wanneer waterstof ontmengt.

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List of Abbreviations and Symbols

Abbreviations

DFT-MD	Density-functional-theory molecular dynamics
ED Table	Extended Data Table of Gilmore & Stixrude (2026)
EOS	Equation of state
MAP	Maximum a posteriori (best-fitting) parameters
RK	Redlich–Kister composition series
xLMA	Extended law of mass action

Symbols

x	H_2 mole fraction on the oxide basis, Eq. (2.6)
y	H_2 mole fraction in native ($MgSiO_3$, H_2) formula units
w	Local hydrogen mass fraction of a closed parcel
g_{mix}	Molar Gibbs energy of mixing per mole of mixing units
$W(x)$	Excess enthalpy at zero pressure, Eq. (2.5)
A, B	Interaction parameters of the published fit
τ, π	Temperature and pressure scales of the excess energy
H_{ex}, S_{ex}, V_{ex}	Excess enthalpy, entropy and volume
T_b	Binodal temperature: first equilibrium separation on cooling
T_{sp}	Spinodal temperature: loss of local stability
ΔT_{ms}	Metastable interval $T_b - T_{\mathrm{sp}}$
f_{H}	Fraction of hydrogen in the H-rich phase at T_{sp}
T_C, x_C	Critical temperature and composition
σ_x, σ_T	Composition and phase-state uncertainties
χ^2	Misfit between model and source data, Eq. (2.11)
R	Gas constant
$R_\oplus$, $M_\oplus$	Earth radius and Earth mass

1. Introduction

1.1 The Sub-Neptune Population and the Characterization Crisis

Sub-Neptune exoplanets represent the most abundant planetary demographic in the Milky Way [7], yet they remain profoundly misunderstood. Defined as planets with radii between Earth ($1\,R_\oplus$) and Neptune ($\approx 4\,R_\oplus$) and masses of 4–10 $M_\oplus$ [7], sub-Neptunes are ubiquitously discovered by transit surveys, including Kepler and TESS. Their frequency alone, surpassing both terrestrial super-Earths and gas giants by a factor of two to three [19], demands that planetary science develop a coherent theoretical framework for understanding their formation, composition, and evolution.

Observational constraints reveal that sub-Neptunes possess a complex internal architecture: rocky or icy cores enveloped by thick, voluminous atmospheres dominated by hydrogen and helium (approximately 75% H and 25% He by mass [18]). This composition places them in a transitional regime between rocky terrestrial planets and gas giants, a structural uniqueness that has no direct analogue in our solar system. This absence of a solar system template proves deeply problematic. We cannot simply scale the physics of terrestrial planet outgassing or giant planet formation to understand sub-Neptunes; their formation and evolutionary pathways must have diverged fundamentally from those of familiar planetary types.

Yet, despite decades of exoplanet discoveries and recent breakthroughs in atmospheric characterization via JWST, fundamental properties remain poorly constrained: the total volatile inventory, the internal redox state of the mantle, the thermal structure at depth, and critically, the composition and long-term stability of the magma ocean boundary.

1.2 The Magma Ocean Problem: Why Coupled Modeling Matters

The defining geophysical feature of young sub-Neptunes is the presence of a **magma ocean**, a region of extensive silicate melting at the top of the planetary mantle, above the core of the planet and directly underlying the volatile-rich atmosphere.

Considerable evidence supports this scenario. The high internal temperatures of sub-Neptunes, maintained by accretionary heating during rapid formation and the insulating effect of their thick H/He envelopes, naturally lead to mantle temperatures exceeding the silicate liquidus at relevant pressures. Magma oceans persist for extended periods, potentially millions of years, as the planet gradually cools and the mantle crystallizes [13]. During this prolonged molten phase, the magma-atmosphere boundary becomes an extraordinarily active chemical interface, not a passive boundary.

Yet the magma ocean problem reveals a critical gap in our theoretical framework. Classical atmosphere models treat the overlying gas as chemically decoupled from the interior, calculating equilibrium compositions as if the atmosphere were isolated from mantle processes. This approach is demonstrably incorrect for sub-Neptunes. In our bachelor thesis [16] we showed with Atmodeller that the magma ocean acts as a dominant control on atmospheric state: a redox-dependent solubility pump stores hydrogen as dissolved water, and the atmospheric CO_2/CH_4 ratio traces the oxidation state of the mantle.

First-principles simulations have since sharpened this picture. Gilmore & Stixrude [9] showed with density-functional-theory molecular dynamics (DFT-MD) that hydrogen and $MgSiO_3$ liquid are **completely miscible** above a critical temperature T_C that decreases from about 3,500 K at 2 GPa to 2,600 K at 10 GPa. The conditions at the magma-atmosphere boundary of most sub-Neptunes lie above this critical curve for much of their early evolution [9, 24]. The interior and the envelope are then one fluid, rather than a melt in contact with a gas. As the planet cools, the interior crosses the coexistence curve (the **binodal**) and hydrogen exsolves into the envelope [21–23]. This is the exsolution of incompatible volatiles during cooling that our thesis identified as a possible cause of late-stage atmospheric inflation [16].

1.3 Spinodal Decomposition and Delayed Exsolution

Current evolution models assume that hydrogen exsolves as soon as the bulk composition enters the miscibility gap, that is, at the binodal [21, 22]. A homogeneous mixture just inside the binodal, however, is *metastable* rather than unstable. Forming the new phase requires nucleation, which has to overcome a free-energy barrier [6, 14]. Only at the **spinodal**, where the curvature of the Gibbs free energy with respect to composition vanishes, does the mixture become unstable to small composition fluctuations and decompose spontaneously. This process is **spinodal**

decomposition [4, 5]. Between the binodal and the spinodal lies a metastable window (Figure 1.1). If nucleation is suppressed, exsolution is delayed until the spinodal. If nucleation is easy, for example on existing interfaces, the delay vanishes.
The size of this window matters for sub-Neptunes, because it sets when, and at which temperature, hydrogen leaves the interior. Delayed exsolution would release hydrogen later and more abruptly. Through the atmospheric hydrogen budget, the timing of this release feeds into the speciation equilibria that set the observable composition, such as the H_2O/H_2 and CO_2/CH_4 ratios that we studied in our thesis [16], and the silane–methane competition proposed as a magma-ocean diagnostic [12]. Whether the window is large enough to matter has not been quantified.
The spinodal is a property of the homogeneous free energy inside the miscibility gap. Simulations of coexisting phases constrain where the gap is, but not the free energy of the homogeneous fluid within it. A spinodal derived from a fitted mixing model therefore inherits the assumptions of its functional form. This makes **identifiability** the central question of this work: how far do the existing data constrain the metastable free energy, and with it the delay?

1.4 Objectives and Scope

This work investigates the thermodynamic limits on hydrogen exsolution from sub-Neptune magma oceans using Atmodeller, a python-based modeling framework. We focus specifically on closed H_2–$MgSiO_3$ parcels with local hydrogen mass fractions ranging from 0.1% to 20%, at pressures of 2 to 10 GPa, cooling through temperatures of 2,000 to 4,000 K. These local mass fractions describe the fluid at the magma-atmosphere boundary, not the hydrogen budget of the whole planet.
The primary objectives of this work are to:

1. **Implement the final 2026 hydrogen–silicate mixing model** [9] in Atmodeller, with a consistent conversion between the published oxide composition basis and the native formula units of Atmodeller.
2. **Quantify the metastable window** between binodal and spinodal along cooling paths, and the hydrogen inventory that delayed exsolution could withhold from separation.
3. **Audit the published fit against its source simulations**, and resolve the mismatches that this audit reveals.
4. **Determine how well existing data constrain the metastable free energy**, by recalibrating several smooth model classes against the same data, and identify the simulations that would settle the question.

This work is constrained to equilibrium thermodynamics and its local stability limits. We acknowledge that real planets may exhibit nucleation on existing interfaces, finite cooling rates, convective mixing and settling of the separated phases. We do not compute a nucleation rate, a planetary radius or an atmospheric spectrum. However, establishing the stability limits is essential: they bound the largest possible delay, define the boundary conditions for kinetic models, and identify which measurements are needed before a kinetic model can be trusted.

2. Methodology

This section outlines the complete modeling framework used to determine the equilibrium and stability limits of hydrogen exsolution at the magma-atmosphere boundary of sub-Neptune exoplanets, called the core–envelope boundary in [9]. The implementation is performed using Atmodeller [2], an open-source Python library designed to resolve coupled chemical equilibrium and mass-balance constraints in gas-melt environments. In the following sections, we outline the conceptual basis, numerical approach, chemical system, and software validation.

2.1 Formulation and Numerical Solution

2.1.1 *Solver Architecture*

Users define the gaseous species, from which the software generates an independent reaction basis using Gaussian elimination (Leal et al. 2016 [17]). At the interface, the solver determines chemical equilibrium among gas species and any specified dissolved species within the melt, subject to elemental mass conservation. The solver determines the equilibrium state by minimizing the total Gibbs free energy using the optimistix framework, with

$$G = \sum_i n_i \mu_i(T, P), \tag{2.1}$$

where n_i is the number of moles and μ_i is the chemical potential of species i, which includes EOS corrections and melt interactions. Atmodeller employs the extended law of mass action (xLMA; Leal et al. 2016 [17]) to build a closed nonlinear system that integrates equilibrium chemistry with elemental mass balance, structured as a root-finding problem, and leverages JAX for high-performance computations (Bradbury et al. 2018 [3]). The nonlinear system

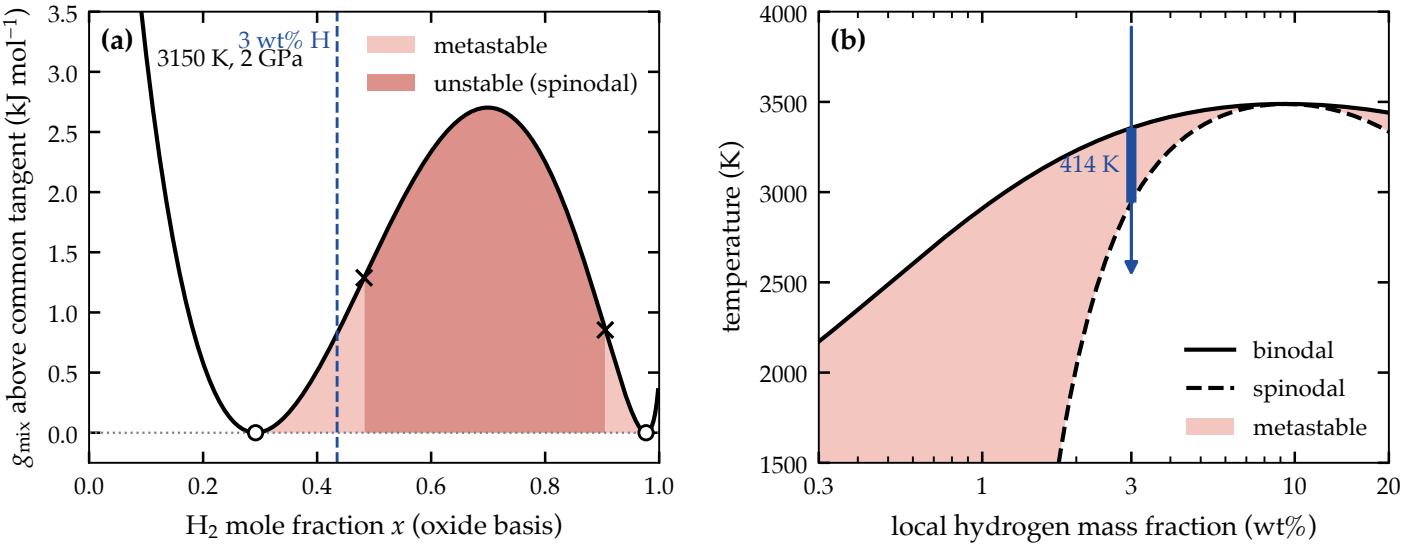


Figure 1.1: ***Binodal, spinodal and the metastable window of a cooling H_2–$MgSiO_3$ parcel.*** *(a) Homogeneous Gibbs energy of mixing at 2 GPa and 3150 K above its common tangent, for the published mixing model [9]. Circles mark the coexisting compositions (binodal), crosses the inflection points (spinodal). A parcel with 3 wt% local hydrogen (dashed line) is metastable here. (b) Binodal and spinodal temperatures at 2 GPa as a function of local hydrogen mass fraction. A 3 wt% parcel cooling at fixed pressure (arrow) enters the miscibility gap at the binodal but becomes unstable only at the spinodal.*

is resolved using robust Newton–type root–finders augmented with line–search and trust–region safeguards.

For this work we extended the phase-equilibrium module of Atmodeller. A binary mixing model now supplies the nonideal activities of two condensed components, here $MgSiO_3$ and H_2. The same Gibbs energy yields the homogeneous activities, the coexisting compositions, the local stability and the equilibrium (convex-envelope) free energy, so that all quantities in this work follow from one function.

2.1.2 *Phase Equilibrium and Local Stability*

For a binary mixture with molar Gibbs energy of mixing $g_{\mathrm{mix}}(x)$, two phases α and β coexist when both component chemical potentials are equal,

$$\mu^{\alpha}_{\mathrm{sil}} = \mu^{\beta}_{\mathrm{sil}}, \qquad \mu^{\alpha}_{\mathrm{H_2}} = \mu^{\beta}_{\mathrm{H_2}}, \tag{2.2}$$

which is the common-tangent construction. The locus of coexisting compositions as a function of temperature is the **binodal**. A homogeneous mixture is locally stable as long as

$$\frac{\partial^2 g_{\mathrm{mix}}}{\partial x^2} > 0, \tag{2.3}$$

and the locus where this curvature vanishes is the **spinodal**. Between the binodal and the spinodal a homogeneous mixture is metastable: it is stable against small composition fluctuations, but a nucleus of the other phase that exceeds a critical size lowers the free energy [6]. Inside the spinodal it decomposes spontaneously [4]. Both conditions are evaluated with bracketed root finding on the two stable branches, which excludes the trivial equal-composition solution, and with implicit derivatives for use inside the Atmodeller solver.

2.2 Hydrogen–Silicate Mixing Model

Gilmore & Stixrude [9] describe their phase-coexistence and homogeneous DFT-MD simulations with an asymmetric regular solution model. We write their equation 8 as

$$\begin{aligned} g_{\mathrm{mix}} = W(x)\left(1 - \frac{T}{\tau} + \frac{P}{\pi}\right) \\ + RT\left[x\ln x + (1-x)\ln(1-x)\right], \end{aligned} \tag{2.4}$$

with the excess enthalpy at zero pressure

$$W(x) = x(1-x)\left[A(1-x) + Bx\right], \tag{2.5}$$

excess entropy W/τ and excess volume W/π. The published parameters are $A = 75\,\mathrm{kJ\,mol^{-1}}$, $B = 424\,\mathrm{kJ\,mol^{-1}}$, $\tau = 4{,}400\,\mathrm{K}$ and $\pi = -35\,\mathrm{GPa}$. The composition variable is the molar fraction of the H_2 component on an oxide basis,

$$\begin{aligned} x &= \frac{N_{\mathrm{H_2}}}{N_{\mathrm{H_2}} + N_{\mathrm{MgO}} + N_{\mathrm{SiO_2}}} \\ &= \frac{N_{\mathrm{H}}/2}{N_{\mathrm{H}}/2 + 2(N_{\mathrm{Mg}} + N_{\mathrm{Si}} + N_{\mathrm{O}})/5}. \end{aligned} \tag{2.6}$$

The paper does not state which component carries A and which carries B. The critical temperature is the same for both choices, but the shape of the miscibility gap is not. Assigning B to the H_2-rich side

(Eq. 2.5) reproduces the published curves of their Figure 2 and their statement that the hydrogen-poor phase holds more hydrogen than the hydrogen-rich phase holds silicate (Section 2.6).

Composition basis. Atmodeller uses native formula units, with the H_2 fraction $y = n_{H_2}/(n_{H_2} + n_{MgSiO_3})$. One $MgSiO_3$ unit contains two oxide units, so

$$x = \frac{y}{2-y}, \qquad G = \left(2\,n_{MgSiO_3} + n_{H_2}\right) g_{mix}(x). \quad (2.7)$$

The second relation keeps the free energy extensive; the native chemical potentials are $\mu_{MgSiO_3} = 2\mu_{oxide}$ and μ_{H_2}. Converting only the endpoint compositions is not sufficient, because the native curvature, activities and phase amounts all depend on the factor $2 - y$. We implemented this conversion as a static multiplicity of the silicate component, available through the constructor `hydrogen_silicate_gilmore26` of the class `SubregularSolution`.

Controls. As controls we retain the historical 2025 parameterization of Rogers, Young & Schlichting [22] on its own formula-unit basis, and a deliberately incorrect import of the 2026 parameters without the basis conversion. The latter is a diagnostic, never a recommended model.

2.3 Delayed-Exsolution Limit

We follow a closed, initially homogeneous parcel that cools at fixed pressure. Its local hydrogen mass fraction w fixes y and hence x. On cooling it first enters the miscibility gap at the binodal temperature $T_b(w, P)$ and later becomes locally unstable at the spinodal temperature $T_{sp}(w, P)$. Because Eq. (2.4) is linear in T, the spinodal temperature follows analytically from Eq. (2.3):

$$T_{sp} = \frac{-W''(x)\,(1 + P/\pi)}{R/[x(1-x)] - W''(x)/\tau}. \quad (2.8)$$

We compare two limiting histories. In the **equilibrium limit** the parcel separates as soon as it crosses T_b. In the **delayed limit** nucleation is suppressed until T_{sp}, after which the parcel equilibrates rapidly. We report the metastable interval

$$\Delta T_{ms} = T_b - T_{sp}, \quad (2.9)$$

and the fraction of all hydrogen that the equilibrium limit already places in the H-rich phase at T_{sp},

$$f_H = \frac{\phi\, y_{rich}}{y_0}, \qquad \phi = \frac{y_0 - y_{poor}}{y_{rich} - y_{poor}}, \quad (2.10)$$

where ϕ is the native mole fraction of the H-rich phase from the lever rule. The quantity f_H is the largest inventory that the delayed limit can withhold from separation. It is a bound between two assumptions, not a computed release rate: both histories conserve all hydrogen, and the delayed parcel still holds its hydrogen dissolved in the silicate.

2.4 Model Configurations and Parameter Space

The results presented in the Results are derived from sweeps over four parameters. When varying a specific parameter, the other three are held at their baseline values unless otherwise noted.

1. **Local Hydrogen Mass Fraction (0.1% – 20%):** This range spans parcels from silicate-rich melt to hydrogen-rich fluid at the magma-atmosphere boundary.
   Baseline: $w = 3$ wt%, a composition that crosses both the binodal and the spinodal above 2,000 K.
2. **Pressure (2 – 10 GPa):** This range covers most of the magma-atmosphere boundary pressures of the sub-Neptune models compiled in [9].
   Baseline: $P = 2$ GPa, the pressure of the best-sampled simulations.
3. **Temperature (2,000 K – 4,000 K):** This window covers the fluid regime of the simulations. We report spinodals below 2,000 K separately and test stopping temperatures of 2,500 and 3,000 K, because neither is a computed melting boundary.
4. **Mixing Model:** the published 2026 fit (baseline), the historical 2025 fit and the incorrect-basis control, and the recalibrated model classes of Section 2.5.

In total the onset grid contains 20,664 rows and the cooling paths 10,827 states. A further 375 deterministic stress cases change A, B, τ and $|\pi|$ by $\pm 10\%$, individually and in combination. These stress cases are not calibrated uncertainties; Section 2.5 replaces them with data-driven posteriors.

2.5 Source-Data Audit and Recalibration

2.5.1 *Source Data*

The source simulations are public [10]. Extended Data (ED) Table 1 lists 14 phase-coexistence runs: four that kept two phases, with the composition of each phase, and ten that became one homogeneous phase, with their bulk composition. The quoted composition uncertainty is ± 0.05 in x. ED Table 2 lists homogeneous simulations at 5 GPa and 4,000 K: three

Table 2.1: ***Model classes used in the recalibration.*** *All classes have constant excess enthalpy, entropy and volume coefficients. The last column counts free parameters.*

Class	Excess Gibbs energy	n
M1	published form, Eq. (2.4)	4
M2	subregular H, S and V, independent	6
M3	RK order 2, published T, P scaling	5
M4	RK order 3, published T, P scaling	6

mixtures and the two endmembers, with enthalpy, entropy and volume per atom. We reconstruct excess properties per mole of mixing units (Eq. 2.6) and remove the ideal mixing entropy from the total entropy. The endmember errors are shared between the three mixtures, so we propagate a full covariance matrix. The endmembers were simulated at 5.00 GPa and the mixtures at 5.06–5.08 GPa; we add the corresponding enthalpy and volume shifts as a fully correlated systematic error. Finally, we digitized the published Figure 2a, which shows the same coexistence runs, from the CC-BY figure file.

2.5.2 *Likelihood and Model Classes*

We compare models with the data through three terms,

$$\chi^2 = \chi^2_{\text{phase}} + \chi^2_{\text{comp}} + \chi^2_{\text{hom}}. \tag{2.11}$$

The first term uses the most robust information in the source: whether each of the 14 runs kept one or two phases. A model that misplaces a run pays $[(T_{\text{run}} - T_b)/\sigma_T]^2$, with T_b the model binodal temperature at the run's bulk composition and $\sigma_T^2 = (50\,\text{K})^2 + (100\,\text{K GPa}^{-1}\,\sigma_P)^2$, which adds the quoted pressure error. The second term compares the eight coexisting compositions, with the quoted ± 0.05 treated as a Gaussian standard deviation. The third term compares the nine homogeneous excess properties with their covariance.

We fit four smooth model classes with constant excess enthalpy, entropy and volume, so that their spinodals follow in closed form. They are written as Redlich–Kister series [20],

$$G_{\text{ex}} = \sum_k (h_k - Ts_k + Pv_k)\, x(1-x)(1-2x)^k, \tag{2.12}$$

summarized in Table 2.1. Class M1 is the published form. M2 frees the shapes of the excess entropy and volume. M3 and M4 add one and two higher composition terms that share the published temperature and pressure scaling.

Priors are uniform within broad stated bounds; we sample $1/\tau$ and $1/\pi$ rather than τ and π. We draw posterior samples with an adaptive Metropolis algorithm [11]: eight chains of 12,000 steps, of which the first 3,000 adapt the proposal and are discarded. We check convergence with the split-chain $\hat{R}$ statistic [8]. Besides the full data set we fit five variants of class M1: with the Figure 2a compositions, with $\sigma_x = 0.15$ for the two runs closest to T_C, without the excess volumes, and with only the phase states plus either the compositions or the homogeneous data. For every retained sample we recompute T_b, T_{sp}, ΔT_{ms}, f_{H} and T_C at the baseline.

2.5.3 *Identifiability Test*

To test whether coexistence data can identify the spinodal at all, we add a nonnegative perturbation that lives only inside the miscibility gap,

$$\delta g = 64\,\lambda B_{\text{gap}}\, u^3(1-u)^3, \quad u = \frac{x - x_{\text{poor}}}{x_{\text{rich}} - x_{\text{poor}}}, \tag{2.13}$$

for $x_{\text{poor}} < x < x_{\text{rich}}$ and zero elsewhere. Here B_{gap} is the height of the published free energy above its common tangent at the middle of the gap. The perturbation and its first two derivatives vanish at the binodal, so all coexisting compositions, all equilibrium free energies and all homogeneous states outside the gap stay exactly the same. The amplitudes $\lambda = 0$ to 8 are a counterexample to uniqueness, not a material model.

2.6 Verification of Thermodynamic Consistency

We verified the implementation in three independent ways. First, our implementation reproduces the published model. Its critical temperatures are 3489 K at 2 GPa and 2643 K at 10 GPa, matching the published 3,500 and 2,600 K, and our binodals follow the digitized curves of the published Figure 2a with an RMS residual of at most 2.3 K (Figure 2.1a). The largest single deviation, 15 K, is about three pixels of the printed line.

Second, a separate scalar implementation, written without any code from Atmodeller, agrees with the Atmodeller binodal and spinodal temperatures to within 10^{-6} K over the grid (Figure 2.1b). The exception is the binodal within 0.02 of x_C, where the flat crest makes the binodal temperature ill-conditioned and the two codes differ by up to 0.11 K. It locates spinodal compositions as the real roots of a polynomial and binodals by nested bracketed root finding, so its errors are independent of the Atmodeller solver.

Third, the Atmodeller test suite passed 288 tests, including the extensive-energy derivative test of the basis conversion (Eq. 2.7), Gibbs–Duhem consistency and spinodal temperatures against independent roots. The study itself checks conservation

of each native component in 27 native equilibrium solves and compares 30 spinodal temperatures with independent SciPy roots. The largest elemental and lever-rule errors are 2.2×10^{-16}.

2.7 Data Processing, Model Limitations and Assumptions

2.7.1 *Data Processing*

Model outputs were exported as CSV and JSON files with SHA-256 hashes of every artifact and analysis script. Data reduction was performed using pandas, and visualizations were generated using matplotlib. Key diagnostics reported in the Results include the binodal and spinodal temperatures, the metastable interval ΔT_{ms}, the withheld hydrogen fraction f_{H} and the critical temperature T_C.

2.7.2 *Model Limitations and Assumptions*

The current framework operates under several constraints. First, the mixture is a binary H_2–$MgSiO_3$ system; helium, iron and other volatiles are absent, and the source simulations used pure $MgSiO_3$. Second, all model classes have constant excess enthalpy, entropy and volume, so the free energy is linear in temperature and pressure. Third, the local mass fractions describe a closed parcel; we do not model transport between the interior and the envelope, convection or settling. Finally, the approach treats nucleation only as two limiting cases, neglecting nucleation rates, interfacial energies, diffusion-limited growth and spatial gradients within the reservoirs.

3. Results

This chapter details the equilibrium and stability limits of hydrogen exsolution from H_2–$MgSiO_3$ mixtures. We begin by characterizing the binodal and spinodal of the published mixing model across pressure and composition, establishing the metastable window. Building on this, we analyze the hydrogen inventory that delayed exsolution could withhold along cooling paths. Subsequently, the published fit is audited against its source simulations and recalibrated against the same data. The chapter concludes with a test of whether coexistence data can identify the spinodal at all.

3.1 Phase Boundaries and the Metastable Window

Figure 3.1 shows the binodal and spinodal temperatures of the published model as a function of local hydrogen mass fraction at 2, 5 and 10 GPa. The binodal is flat-topped: its crest at 2 GPa lies at $T_C = 3489$ K and $x_C = 0.720$ ($w = 9.36$ wt%), and it falls only slowly away from x_C. The spinodal touches the binodal at the critical point and falls steeply on the hydrogen-poor side. Spinodals exist only above $w = 1.40$ wt%. Below this threshold a parcel can still separate in equilibrium, but its homogeneous state never becomes unstable, so separation always requires nucleation (Appendix A).

Table 3.1: ***Onset temperatures for a parcel with 3 wt% local hydrogen.*** *f_{H} is the fraction of hydrogen that equilibrium places in the H-rich phase at the spinodal.*

Model, P	T_b (K)	T_{sp} (K)	ΔT_{ms} (K)	f_{H} (%)
2026 fit, 2 GPa	3356	2942	414	65.5
2026 fit, 5 GPa	3051	2675	376	65.5
2026 fit, 10 GPa	2542	2229	313	65.5
2025 fit, 2 GPa	3962	3908	53	39.0
Wrong basis, 2 GPa	3468	3422	46	36.1

For the baseline parcel of 3 wt% hydrogen at 2 GPa, the binodal lies at 3356 K and the spinodal at 2942 K. The metastable interval is therefore 414 K (Table 3.1). It shrinks to 376 K at 5 GPa and 313 K at 10 GPa, because both onset temperatures scale with $1 + P/\pi$. At 10 GPa the spinodal lies below the coldest two-phase simulation (2,500 K), so it is an extrapolation.

The composition basis matters. The historical 2025 parameterization gives an interval of only 53 K, and importing the 2026 parameters without the basis conversion gives 46 K. With the correct conversion the interval is about nine times larger. Changing A, B, τ and $|\pi|$ by $\pm 10\%$ gives intervals of 329–504 K. These stress cases are not calibrated uncertainties; Section 3.4 derives uncertainties from the data.

3.2 Hydrogen Inventory along Cooling Paths

Figure 3.2a follows the 3 wt% parcel as it cools. In the equilibrium limit the H-rich phase appears at T_b and grows on further cooling. In the delayed limit nothing separates until T_{sp}, where the parcel jumps to the equilibrium state. At the spinodal the equilibrium limit already places 65.5% of all hydrogen in the H-rich phase, while the delayed limit still holds it in the homogeneous parcel.

At 3,000 K and 2 GPa the baseline parcel is metastable. Equilibrium would place 61.4% of its hydrogen in the H-rich phase, which is 18.4 g of the 30 g of hydrogen in each kilogram of parcel, whereas the delayed limit has not separated at all. Equilibration would lower the Gibbs energy by 2.98 kJ per native mole. This is a free-energy difference, not a prediction of released heat.

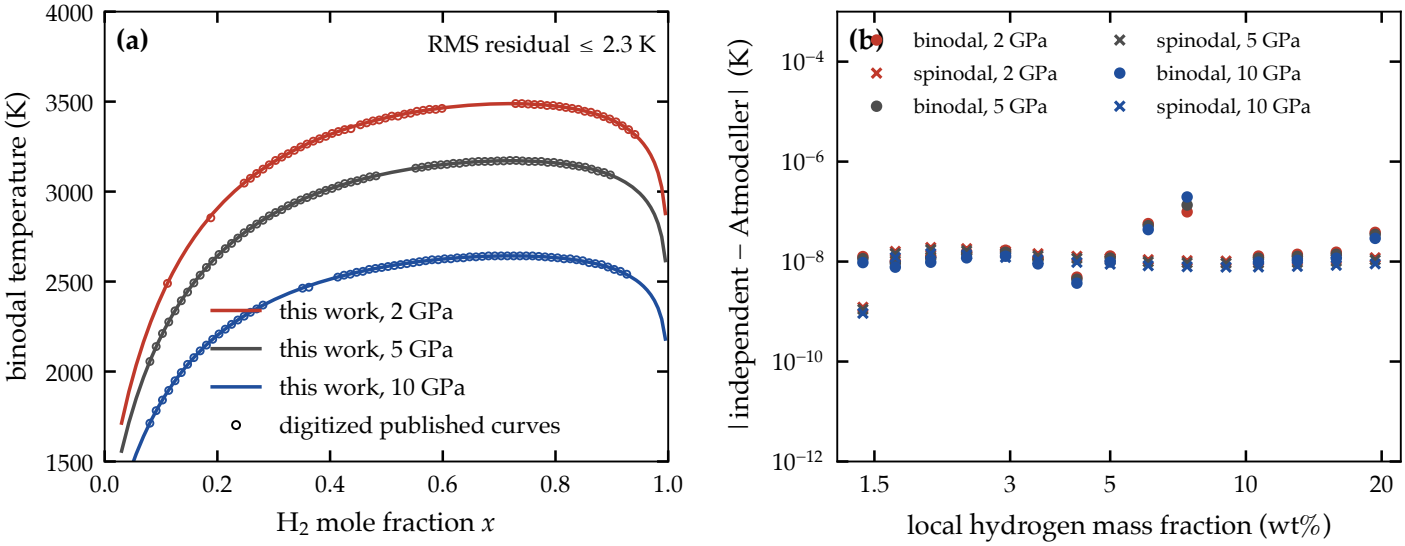


Figure 2.1: ***Internal consistency check of Atmodeller.*** *(a) Binodal temperatures of our implementation of Eq. (2.4) (lines) and the curves digitized from Figure 2a of [9] (circles) at 2, 5 and 10 GPa. (b) Absolute difference between a separate scalar implementation and Atmodeller for the binodal (circles) and spinodal (crosses) temperatures across the onset grid. The two codes share no solver code.*

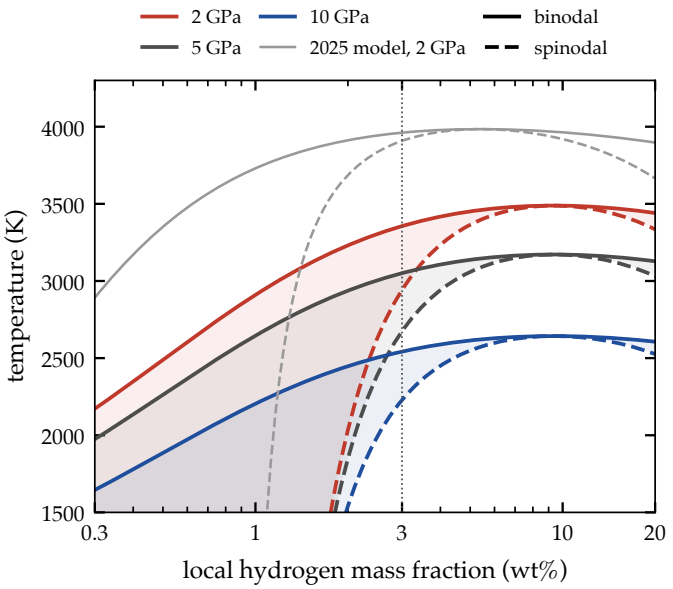


Figure 3.1: ***Binodal and spinodal temperatures of the published mixing model.*** *Solid lines show the binodal, dashed lines the spinodal, at 2, 5 and 10 GPa; shading marks the metastable window. Grey lines show the historical 2025 parameterization at 2 GPa. The dotted line marks the 3 wt% baseline.*

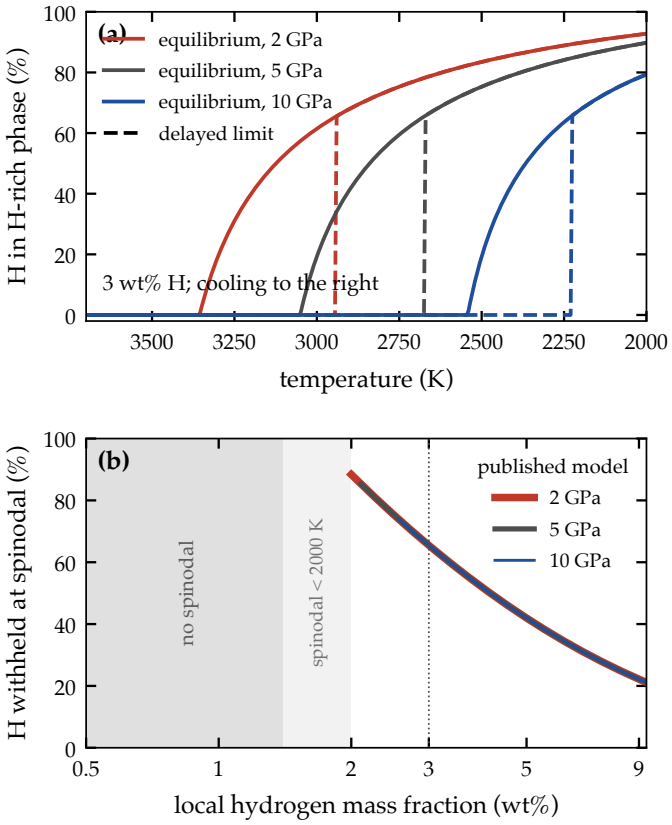


Figure 3.2: ***Hydrogen withheld by delayed exsolution.*** *(a) Fraction of all hydrogen in the H-rich phase for a 3 wt% parcel cooling at 2, 5 and 10 GPa, in the equilibrium limit (solid) and the delayed limit (dashed). (b) Fraction withheld at the spinodal as a function of local hydrogen mass fraction. The curves for the three pressures coincide.*

The withheld fraction at the spinodal is the same at 2, 5 and 10 GPa to within 2×10^{-10}, or 7×10^{-6} within 0.1 wt% of the critical composition, where the lever rule becomes ill-conditioned (Figure 3.2b). It falls from 88% at 2 wt%, the lowest composition whose spinodal lies above 2,000 K, to about 21% near the critical composition. The pressure invariance follows from the separable form of Eq. (2.4) (Appendix A); it is not independent evidence.

3.3 Audit of the Published Fit against its Source Data

Our implementation reproduces the published model (Section 2.6), so any mismatch with the source

simulations belongs to the published fit itself. The fit agrees with all ten one-phase runs. For the two-phase run at 1.65 GPa, its gap at 3,500 K lies entirely above the bulk composition of the cell, a miss of 35 K in temperature. It reproduces both compositions within the quoted ± 0.05 for only one of the four two-phase runs (Table B.1, Figure 3.3b).

Near-critical runs. The largest misses occur for the two runs closest to the critical curve. The published T_C lies 26 K above the run at 1.65 GPa and 3,500 K, and 157 K above the run at 5.14 GPa and 3,000 K. For the first run the model predicts a gap centred at $x \approx 0.71$, whereas the simulation shows a gap centred at 0.52. For exactly these two runs the source table and the source figure disagree. The digitized Figure 2a places the pairs at (0.419, 0.629) and (0.514, 0.736), whereas ED Table 1 lists (0.458, 0.576) and (0.587, 0.706). The differences range from -0.073 to $+0.053$. For the two wide-gap runs, table and figure agree to within 0.0006.
In all four two-phase runs, the per-phase atom numbers in ED Table 1 add up to the full simulation cell, so every interfacial atom is counted in one of the two phases. With sharp interfaces this matters little. Near T_C, however, interfaces broaden, and such a count pulls both compositions toward the bulk composition of the cell, which is 0.526 and 0.629 for these runs. The two composition estimators described in the source methods [9] could therefore differ most for exactly these runs, as observed. We treat the near-critical compositions as uncertain at the 0.04–0.07 level. This is an inference from the published numbers, not a demonstrated cause.

Homogeneous simulations. The homogeneous simulations reveal a second mismatch (Figure 3.3c–e). The published fit predicts an excess volume of $-1.94\,\mathrm{cm^3\,mol^{-1}}$ at $x = 0.588$, whereas the simulations give $-1.00 \pm 0.17\,\mathrm{cm^3\,mol^{-1}}$. Because the excess volume sets the pressure dependence of T_C, the value $\pi = -35$ GPa reproduces the critical curve between 2 and 10 GPa, but the directly simulated volumes imply a pressure scale about twice as large in magnitude. The fit also underestimates the excess enthalpy at $x = 0.24$. In total, the published fit has $\chi^2 = 0.4$ for the 14 phase states, 71.3 for the eight compositions and 111.4 for the nine homogeneous properties.
These comparisons reproduce checks against the data from which the fit was built. They are not independent validation, and they do not show that any simulation is wrong. They show that one four-parameter model cannot follow all of the source data at once.

3.4 Recalibrated Delayed-Exsolution Interval

Table 3.2 and Figure 3.4 summarize the nine refits. Refitting the published form (class M1) to all data lowers χ^2 from 183 to 54. The best fit is much less asymmetric than the published one (Table C.1) and has a lower critical temperature, $T_C(2\,\mathrm{GPa}) = 3235$ K (68% range 3212–3259 K). It pays for this with the phase state of the run at 1.65 GPa and 3,500 K, which it places 249 K above its binodal. The metastable interval of the baseline parcel shrinks to 90 K (68–115 K), and the withheld fraction to 48%.
The data treatments show which data drive this shift (Table C.1). Without the homogeneous data, the fit returns to the published asymmetry, $B/A = 6.1$ against 5.65, but with interaction parameters three times larger and a lower τ. Its interval is 132 K, not 414 K, so the asymmetry alone does not set the interval: the excess entropy and the strength of the interaction matter as much. With the homogeneous data but without the compositions, the excess is nearly symmetric, $B/A = 1.4$, and the interval is 65 K. The compositions themselves discriminate weakly between these fits: their misfit is $\chi^2_{\mathrm{comp}} = 11.7$ for the strongly and 12.2 for the weakly asymmetric fit. The difference lies mainly in the phase states and the homogeneous data. Using the Figure 2a compositions instead of ED Table 1 gives 92 K, and dropping the excess volumes gives 86 K: the volume mismatch controls the pressure dependence of T_C, but hardly the interval at 2 GPa. Giving the two near-critical runs a composition uncertainty of 0.15 raises the asymmetry to $B/A = 2.4$ and lengthens the interval to 220 K.
Every refit that uses the coexisting compositions places the run at 1.65 GPa and 3,500 K between 100 and 257 K above its binodal. This reflects a tension within the source data. The narrow gap at 5.14 GPa and 3,000 K requires T_C close to 3,000 K at that pressure, whereas the two-phase state at 1.65 GPa requires $T_C > 3{,}500$ K. A linear critical curve meets both only if it falls by more than about 143 K per GPa. The published fit falls by 106 K per GPa, and the fit to phase states and homogeneous data by 54 K per GPa.
The more flexible classes fit the data with $\chi^2 = 45$ (M2), 49 (M3) and 48 (M4), against 54 for M1. Their intervals are 42, 87 and 144 K. None of the classes fits all data within the quoted errors: for 31 data points, χ^2 stays between 45 and 54. Across all nine refits the median interval ranges from 42 to 220 K, compared with 414 K for the published fit (Table 3.2, Figure 3.4a). The most flexible class, M4, does not

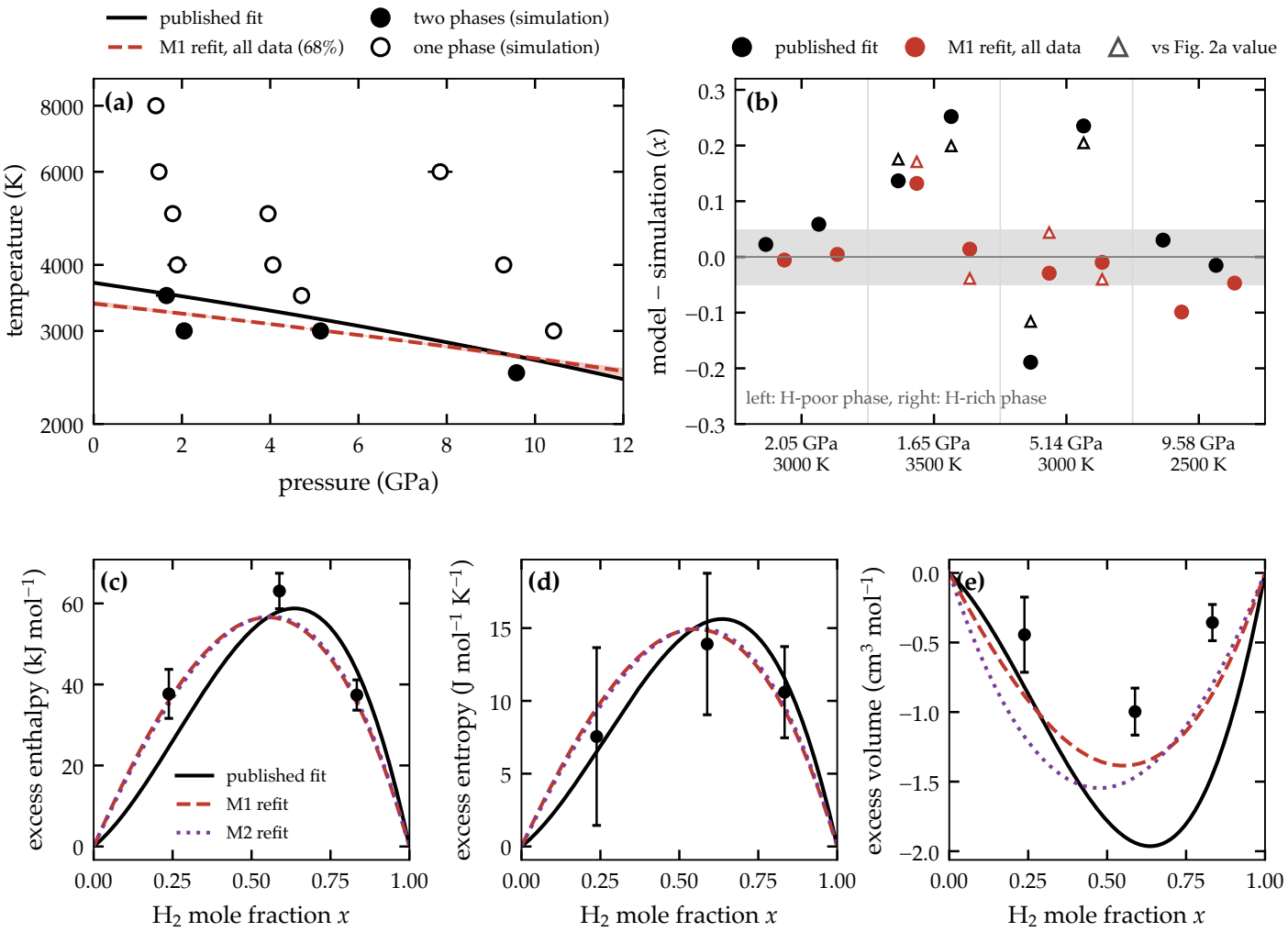


Figure 3.3: ***Audit of the published fit against its source simulations.*** *(a) Phase state of the 14 coexistence runs (filled: two phases; open: one phase) with the published critical curve (black) and the critical curve of the refit to all data (red, 68% band). (b) Model minus simulation for the eight coexisting compositions, against ED Table 1 (circles) and the digitized Figure 2a (triangles); the grey band is the quoted ±0.05. (c–e) Excess enthalpy, entropy and volume of the homogeneous simulations at 5 GPa and 4,000 K (symbols, 1σ), with the published fit and the refits of classes M1 and M2.*

exclude the published value: its 95% range is 59–479 K. Every fit met our convergence criterion of a split-chain $\hat{R} < 1.1$; the largest value is 1.05.

3.5 Identifiability of the Spinodal

The perturbation of Eq. (2.13) leaves the binodal, the equilibrium free energy and every homogeneous state outside the gap unchanged; we verified this to within $4 \times 10^{-12}\,\mathrm{J\,mol^{-1}}$. It nevertheless moves the spinodal (Figure 3.6). At $\lambda = 8$ the 3 wt% spinodal rises by 35.9 K, the 5 wt% spinodal falls by 13.5 K and the 2 wt% spinodal rises by 250.4 K. Coexistence data alone therefore cannot identify the spinodal. Any spinodal inferred from them is a property of the assumed functional form.

4. Discussion

The Results reveal that the onset of hydrogen exsolution from a sub-Neptune magma ocean cannot be read from equilibrium miscibility data alone. The published mixing model predicts a metastable window of several hundred kelvin, but the same source data, fitted with the same or slightly more flexible forms, favour windows two to ten times shorter. In this chapter, we interpret the physical mechanisms behind these results and discuss how they bear on the evolution and observable properties of sub-Neptune exoplanets.

4.1 Why Equilibrium Data Cannot Fix the Spinodal

The binodal and the spinodal probe different parts of the free energy. The binodal follows from the com-

Table 3.2: ***Recalibration of the hydrogen–silicate mixing model.*** *χ^2 terms at the best fit for the 14 phase states, the 8 compositions and the 9 homogeneous properties, and posterior medians with 68% ranges of the critical temperature at 2 GPa, the metastable interval and the withheld hydrogen fraction for a 3 wt% parcel at 2 GPa. Fits marked † leave out data (dashes), so their χ^2 cannot be compared with the others.*

Fit	χ^2_{phase}	χ^2_{comp}	χ^2_{hom}	T_C (K)	ΔT_{ms} (K)	f_{H} (%)
Published fit	0.4	71.3	111.4	3489	414	65
M1, all data	20.2	12.2	21.9	3235 [3212–3259]	90 [68–115]	48 [45–51]
M1, Fig. 2a compositions	17.9	18.5	21.8	3255 [3228–3281]	92 [69–122]	49 [45–52]
M1, $\sigma_x = 0.15$ near T_C	3.3	17.8	19.1	3412 [3347–3477]	220 [157–295]	58 [54–61]
M1, without volumes†	10.4	13.8	2.3	3298 [3276–3318]	86 [64–111]	49 [46–52]
M1, phases + compositions†	9.0	11.7	–	3311 [3289–3328]	132 [83–198]	65 [60–68]
M1, phases + homogeneous†	0.1	–	10.0	3480 [3425–3553]	65 [23–120]	44 [36–51]
M2, all data	21.5	5.0	18.8	3221 [3196–3248]	42 [24–68]	41 [36–46]
M3, all data	20.1	13.4	15.1	3244 [3217–3277]	87 [67–111]	41 [39–44]
M4, all data	20.0	15.1	13.4	3265 [3230–3319]	144 [92–248]	47 [42–54]

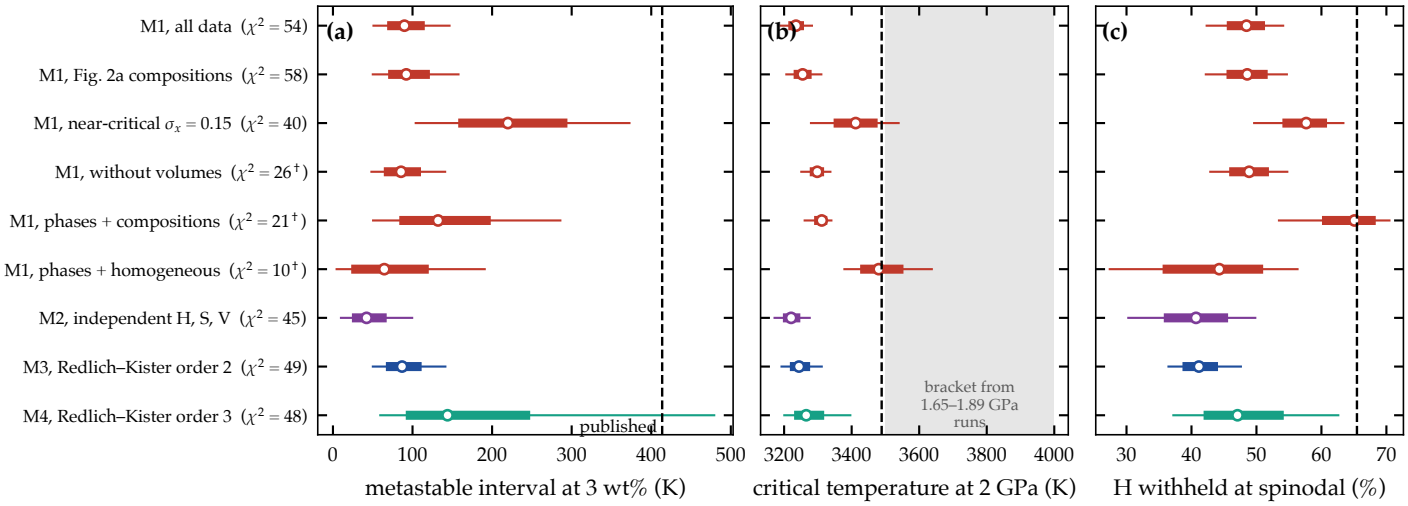


Figure 3.4: ***Posterior distributions of the delayed-exsolution quantities.*** *(a) Metastable interval, (b) critical temperature at 2 GPa and (c) withheld hydrogen fraction for a 3 wt% parcel at 2 GPa, for each model class and data treatment. Thick bars show 68% and thin bars 95% intervals; open circles mark medians. Dashed lines mark the published fit; the grey band in (b) is the bracket set by the two-phase run at 1.65 GPa and the one-phase run at 1.89 GPa, drawn without correction to 2 GPa. Labels give the best-fit χ^2; fits marked † use fewer data.*

mon tangent, so it depends only on the free energy at the two coexisting compositions and on the line between them. The spinodal depends on the local curvature of the homogeneous free energy between those compositions, where a phase-separated simulation never samples the homogeneous state. The perturbation of Section 3.5 makes this explicit: it changes the curvature inside the gap and leaves every equilibrium observable unchanged.

Fitting a smooth model with few parameters ties the curvature inside the gap to the free energy outside it. The spinodal is then exactly as well constrained as these parameters. For the baseline parcel three of them matter together: the asymmetry B/A, which sets the critical composition, the excess entropy, set by τ, and the strength of the interaction. The source data pull them in different directions. Without the homogeneous data, the fit returns to the published asymmetry ($B/A = 6.1$), but with a larger excess entropy than the homogeneous simulations allow. The homogeneous enthalpies, which are almost symmetric about $x \approx 0.54$, favour a weak asymmetry ($B/A = 1.4$). The published fit combines the strong asymmetry with moderate interaction parameters. None of our refits reproduces this combination, and none has a median window above 220 K. A weaker asymmetry moves the critical composition toward the baseline parcel and shortens its window. A second tension concerns the critical curve: its slope, set by the excess volume in all our model classes, cannot follow both the near-critical runs and the homogeneous volumes (Section 3.4).

Close to the critical point, binodal and spinodal are linked more tightly. In mean-field theory, to leading order, the spinodal lies three times as far below T_C as the binodal at the same distance from x_C [6]. Near-

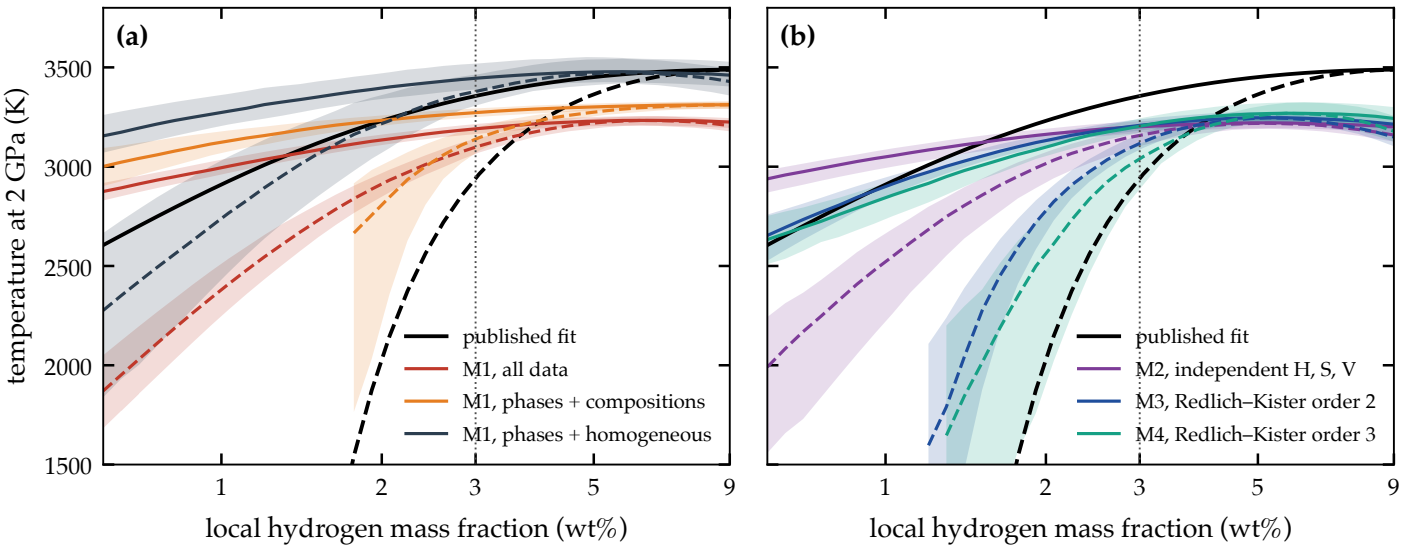


Figure 3.5: ***Posterior binodals and spinodals at 2 GPa.*** *Medians (lines) and 68% bands of the binodal (solid) and spinodal (dashed) for (a) class M1 with three data treatments and (b) classes M2–M4 with all data, compared with the published fit (black).*

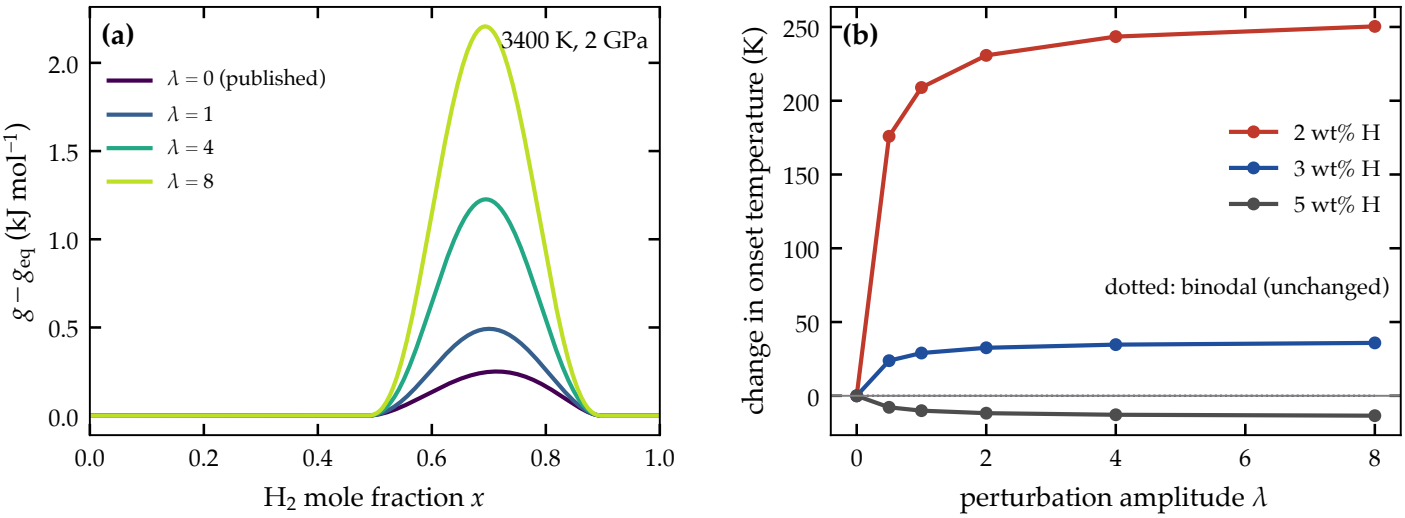


Figure 3.6: ***Two free energies with the same equilibrium but different spinodals.*** *(a) Homogeneous Gibbs energy above the equilibrium (convex-envelope) free energy at 3,400 K and 2 GPa for perturbation amplitudes λ = 0 to 8. Outside the gap all curves are zero. (b) Change of the spinodal (solid) and binodal (dotted) temperatures with λ for 2, 3 and 5 wt% local hydrogen.*

critical data therefore constrain the spinodal of near-critical parcels, but not of a parcel at $x = 0.435$ that lies far from x_C. The runs that are most informative about the curvature are also the runs whose compositions are least certain (Section 3.3).

4.2 Implications for Hydrogen Release and Atmospheric Composition

Evolution models with hydrogen–silicate miscibility release hydrogen into the envelope as the binodal moves through the interior [21–23]. They assume that separation starts at the binodal. Our results bound the possible lag. For a parcel with 3 wt% local hydrogen at 2 GPa, nucleation-limited release could lag the binodal by 42 to 414 K in temperature, depending on which description of the source data is adopted. Over this interval, equilibrium would already have moved up to 65.5% of the parcel's hydrogen into a separate H-rich phase.

A lag in temperature becomes a lag in time through the cooling rate of the planet at that stage, which we do not compute. If it is long, it postpones the inflation of the envelope by exsolved hydrogen and makes the release more abrupt once the spinodal is reached. In our thesis, the atmospheric hydrogen budget controlled the mean molecular weight μ_{atm} and the carbon speciation through the methanation equilibrium $CO_2 + 4H_2 \rightleftharpoons CH_4 + 2H_2O$ [16]. A delayed, abrupt supply of hydrogen would temporarily lower μ_{atm}, shift carbon toward CH_4, and could change the silane–methane competition proposed as a magma-ocean diagnostic [12]. These effects remain conditional on the transport of hydro-

gen from the interior to the observable atmosphere, which neither this work nor the equilibrium models resolve. In an earlier sensitivity experiment with the 2025 mixing model (repository directory `research/spinodal_study`), the atmospheric signal of demixing depended strongly on the assumed transfer of material and on condensation.

Whether the delayed limit applies at all depends on nucleation. Interfaces lower or remove the nucleation barrier: the magma-atmosphere boundary itself, crystals in a cooling magma ocean and droplets of an earlier-separated phase can all act as nucleation sites. Deep in a supercritical interior without such interfaces, separation would instead require homogeneous nucleation. The metastable window is therefore an upper bound on the lag, and the equilibrium binodal a lower bound.

4.3 A Decisive Simulation Campaign

The posteriors identify which simulations would settle the question. A screen over candidate measurements uses the posterior samples of all four model classes with equal weight (Figure 4.1). One new homogeneous simulation at $x \approx 0.35$, with the precision of the existing ED Table 2 entries, would reduce the posterior standard deviation of the baseline spinodal only from 83 K to 75 K. Most of the remaining spread comes from the differences between model classes and between data subsets, which one more homogeneous point does not remove.

A direct measurement of the curvature is more powerful. At 2 GPa and 3200 K, a 10% measurement of $\partial^2 g/\partial x^2$ would reduce the standard deviation to 25 K. At this temperature 95% of the posterior samples leave the baseline parcel locally stable: 46% metastable and 48% still one phase. The curvature is accessible without a free-energy model, from the long-wavelength limit of the Bhatia–Thornton concentration–concentration structure factor, $S_{cc}(0) = RT/(\partial^2 g/\partial x^2)$ per mole of mixing units [1, 15]. Homogeneous DFT-MD simulations of the metastable parcel, in cells large enough to sample composition fluctuations over several nanometres, could measure it. Simulations below T_C would also test the assumption of temperature-independent excess enthalpy and entropy, which all four model classes share.

Even a measured spinodal fixes only the limit of the delay. Its actual length requires the interfacial tension and the mobility of the mixture. The interface fluctuations in the existing coexistence simulations could constrain the tension, because their magnitude is related to it [9]. Together they give nucleation rates through classical nucleation theory [14], which an evolution model can then use in place of the two limiting cases studied here.

5. Conclusion

The study of sub-Neptune exoplanets occupies a critical intersection between geophysics and atmospheric science. This work has demonstrated that the release of hydrogen from a miscible sub-Neptune interior cannot be timed from equilibrium miscibility data alone. Between the equilibrium onset of exsolution and the loss of local stability lies a metastable window, and its width is set by a part of the free energy that the existing simulations do not sample. By utilizing the Atmodeller framework to compute binodals, spinodals and their uncertainties for H_2–$MgSiO_3$ parcels across local hydrogen mass fractions of 0.1–20% and pressures of 2–10 GPa, we have constrained the thermodynamic limits on hydrogen exsolution during the cooling of the magma ocean. This chapter summarizes the primary insights derived from this modeling and discusses their implications for interpreting observations.

5.1 Summary of Findings

Our modeling yields four primary conclusions regarding the exsolution of hydrogen from sub-Neptune interiors.

5.1.1 A Composition-Consistent Mixing Model

We implemented the final 2026 hydrogen–silicate mixing model in Atmodeller. The published model uses oxide units, whereas Atmodeller uses $MgSiO_3$ formula units, and the conversion must act on the extensive free energy, not only on the compositions. With this conversion our implementation reproduces the published phase diagram with an RMS residual of 2.3 K. Without it, the metastable interval of the baseline parcel is about nine times too short.

5.1.2 The Metastable Window

With the published fit, a parcel with 3 wt% local hydrogen at 2 GPa enters the miscibility gap at 3356 K and becomes unstable at 2942 K. Over this 414 K interval, delayed exsolution could withhold up to 65.5% of its hydrogen from separation. In the published fit, parcels with less than 1.40 wt% hydrogen never become unstable, so their separation always requires nucleation; in the refits this threshold lies lower.

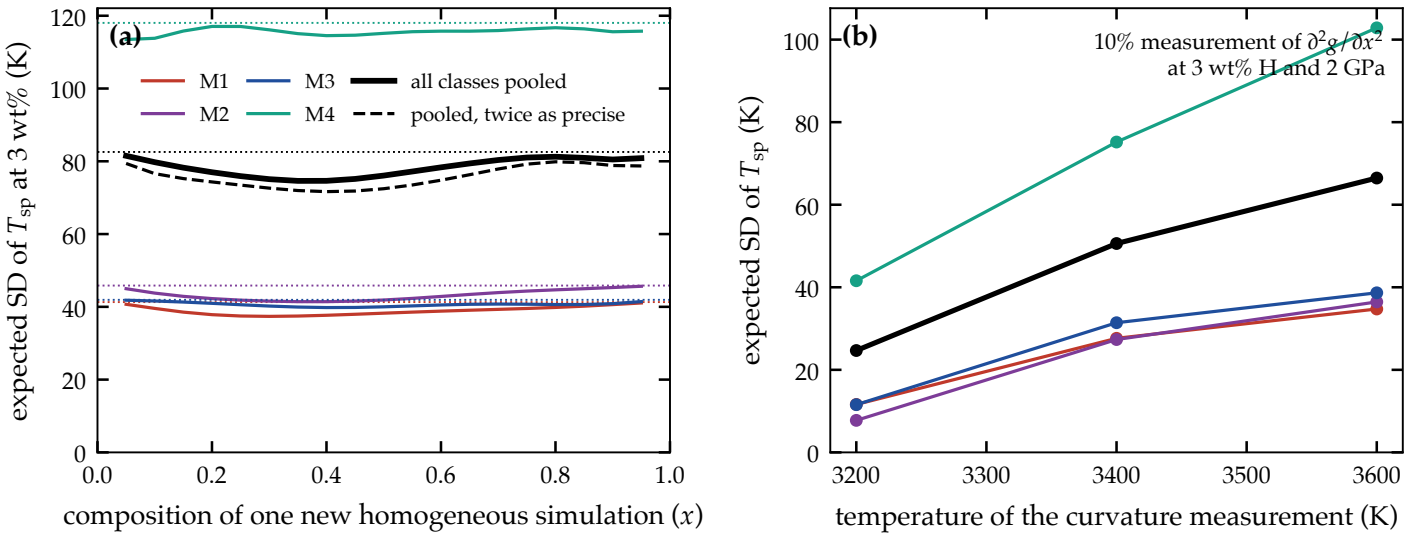


Figure 4.1: ***Simulations that would constrain the spinodal.*** *Expected posterior standard deviation of the spinodal temperature of a 3 wt% parcel at 2 GPa after (a) one new homogeneous simulation of excess enthalpy, entropy and volume at composition x, for each model class (thin lines) and all classes pooled (thick line; dotted lines: current values), and (b) a measurement of the composition curvature $\partial^2 g/\partial x^2$ at the baseline composition with 10% precision, at three temperatures. Linear-Gaussian screen of the recalibration samples.*

5.1.3 The Source Data Do Not Fix the Window

The published fit departs from its own source simulations in two places: the compositions of the two runs closest to the critical curve, where the source table and figure also disagree, and the excess volume, which is about half the value that the fitted pressure scale implies. Refitting the same model form to all source data shortens the interval to 90 K (68% range 68–115 K). Across four smooth model classes and six data treatments, the median interval ranges from 42 to 220 K, and the most flexible class still allows the published value. The window depends on the asymmetry, the excess entropy and the strength of the interaction together, and the coexisting compositions and the homogeneous simulations pull these in different directions.

5.1.4 The Spinodal Is Not Identifiable from Coexistence Data

A free-energy perturbation that leaves every equilibrium observable unchanged still moves the spinodal by tens to hundreds of kelvin. Any spinodal inferred from coexistence data is therefore a property of the assumed functional form. The curvature inside the miscibility gap has to be measured, not inferred.

5.2 Implications for Observations

These findings introduce a new uncertainty into the interpretation of young sub-Neptunes. Evolution models with hydrogen–silicate miscibility time the release of hydrogen with the equilibrium binodal. If nucleation is suppressed, the release can lag by up to several hundred kelvin of cooling and then occur abruptly.

Therefore, observational constraints from instruments such as JWST should be interpreted within a coupled framework:

1. **Radius evolution:** Inflation of the envelope by exsolved hydrogen may set in later than equilibrium models predict. The demographic tests proposed for young sub-Neptunes [21] should therefore include the metastable window as a systematic uncertainty.
2. **Atmospheric composition:** A delayed and abrupt supply of hydrogen would temporarily lower the mean molecular weight and shift the carbon speciation toward CH_4, in line with the hydrogen-budget dependence found in our thesis [16]. Such a signal requires efficient transport from the interior and remains untested.
3. **Interpretive caution:** Because the window is not identifiable from present data, atmospheric or radius signatures should not yet be interpreted as evidence for or against spinodal decomposition.

5.3 Future Outlook

While this work establishes a thermodynamic bound on delayed hydrogen exsolution, the sub-Neptune environment remains complex. The current models assume a binary H_2–$MgSiO_3$ mixture with temperature-independent excess enthalpy and entropy. Future simulations should measure the curvature of the homogeneous free energy inside the mis-

cibility gap directly, through the concentration structure factor, and test the temperature dependence of the excess properties below T_C. The interfacial tension and mobility then fix nucleation rates, which an evolution model can use instead of the two limiting cases studied here. Lastly, as the magma ocean solidifies, the exsolution of incompatible volatiles could lead to late-stage atmospheric inflation. This, too, should be incorporated as a time-dependent evolution of the cooling and crystallizing magma ocean, coupled to the atmospheric chemistry of Atmodeller. Despite these open questions, this work highlights that accurately predicting when a sub-Neptune releases its hydrogen requires a fundamental understanding of the metastable states of its interior.

Appendices

A. Thermodynamics of the Mixing Model

This appendix presents a self-contained theoretical framework for the hydrogen–silicate mixing model used in `Atmodeller`. It unifies the free energy, the composition basis, the chemical potentials and the stability conditions. The goal is to provide a physically transparent explanation of why the binodal and the spinodal of the published model behave as they do, and of which parts of that behaviour come from the model form rather than from the data.

A.1 Chemical Potentials on the Oxide Basis

For a binary mixture of oxide units (component 1) and H_2 (component 2), with $g_{\mathrm{mix}}(x)$ per mole of mixing units and x the H_2 fraction, the chemical potentials relative to the pure components are

$$\mu_1 = g_{\mathrm{mix}} - x\, g'_{\mathrm{mix}}, \qquad \mu_2 = g_{\mathrm{mix}} + (1-x)\, g'_{\mathrm{mix}}. \tag{A.1}$$

For Eq. (2.4) with $f = 1 - T/\tau + P/\pi$, the excess parts are

$$\mu_1^{\mathrm{ex}} = f\, x^2\, [B + 2(A-B)(1-x)]\,, \tag{A.2}$$
$$\mu_2^{\mathrm{ex}} = f\, (1-x)^2\, [A + 2(B-A)x]\,, \tag{A.3}$$

which is equation 14 of [9] written for both components.

A.2 Conversion to Native Formula Units

`Atmodeller` counts $n_{\mathrm{MgSiO_3}}$ and $n_{\mathrm{H_2}}$. With $G = (2n_{\mathrm{MgSiO_3}} + n_{\mathrm{H_2}})\, g_{\mathrm{mix}}(x)$ and $x = n_{\mathrm{H_2}}/(2n_{\mathrm{MgSiO_3}} + n_{\mathrm{H_2}})$,

$$\frac{\partial G}{\partial n_{\mathrm{MgSiO_3}}} = 2\mu_1, \qquad \frac{\partial G}{\partial n_{\mathrm{H_2}}} = \mu_2. \tag{A.4}$$

Per native mole, $h(y) = (2-y)\, g_{\mathrm{mix}}(x(y))$ with $x = y/(2-y)$ and $dx/dy = 2/(2-y)^2$. Differentiating twice gives

$$h''(y) = \frac{4\, g''_{\mathrm{mix}}(x)}{(2-y)^3}. \tag{A.5}$$

The curvature changes magnitude but not sign, so the spinodal is the same in both bases, as it must be. The coexisting compositions are also the same once converted with Eq. (2.7). Importing the published parameters without the factor $2-y$, as in our incorrect-basis control, changes the free energy itself and therefore moves both curves.

A.3 Spinodal Temperature and Its Threshold

The excess curvature of Eq. (2.5) is linear in composition,

$$W''(x) = 2B - 4A + 6(A-B)\, x. \tag{A.6}$$

Setting $g''_{\mathrm{mix}} = 0$ gives Eq. (2.8). A positive spinodal temperature requires $W''(x) < 0$, that is

$$x > x_{\min} = \frac{2B - 4A}{6(B-A)} = 0.262 \tag{A.7}$$

for the published parameters. This corresponds to $w = 1.40$ wt% local hydrogen. Below this threshold a parcel can still separate in equilibrium, but its homogeneous state never becomes unstable: separation there always requires nucleation.

A.4 Why the Withheld Fraction Does Not Depend on Pressure

Write $g_{\mathrm{mix}}/(RT) = s(x) + r\,W(x)$, with s the ideal term and

$$r(T,P) = \frac{1 - T/\tau + P/\pi}{RT}. \tag{A.8}$$

Phase equilibrium and stability depend on T and P only through r. At fixed composition the spinodal fixes $r_{\mathrm{sp}} = -1/[x(1-x)W''(x)]$, and the binodal fixes another value $r_b(x)$. Solving $r(T,P) = r_*$ gives

$$T_*(x,P) = \frac{1 + P/\pi}{R\,r_* + 1/\tau} = T_*(x,0)\left(1 + \frac{P}{\pi}\right). \tag{A.9}$$

Both onset temperatures therefore scale with $1 + P/\pi$, and the coexisting compositions at the spinodal, hence f_{H}, are the same at every pressure. The numerical invariance of f_{H} (Section 3.2) is a consequence of this separable form, not independent evidence. Classes M3 and M4 share it, because all their terms scale with the same factor; class M2 does not.

A.5 Critical Point

At the critical point both the second and third derivatives of g_{mix} vanish. For Eq. (2.4) this gives equations 17–19 of [9]: the ratio B/A fixes the critical composition, here $x_C = 0.720$ ($w = 9.36$ wt%), and

$$T_C(P) = T_C^0\left(1 + \frac{P}{\pi}\right), \qquad T_C^0 = 3700\,\mathrm{K}. \tag{A.10}$$

The critical temperature is the same whichever component carries A; the critical composition is not.

A.6 Curvature Change of the Identifiability Perturbation

For the perturbation of Eq. (2.13) with gap width $\Delta x = x_{\mathrm{rich}} - x_{\mathrm{poor}}$,

$$\frac{\partial^2 \delta g}{\partial x^2} = \lambda B_{\mathrm{gap}}\,\frac{384\,u(1-u)(1-5u+5u^2)}{\Delta x^2}. \tag{A.11}$$

The change is negative in the middle of the gap ($0.28 < u < 0.72$) and positive near its edges, so the perturbation can move the spinodal in either direction. This is why the 3 wt% spinodal rises while the 5 wt% spinodal falls in Figure 3.6. The value, first and second derivatives of δg vanish at $u = 0$ and $u = 1$, which keeps every equilibrium quantity unchanged.

B. Source Data and Audit Tables

This appendix reproduces the source simulations of Gilmore & Stixrude [9] that enter the audit and the recalibration, next to the predictions of the published fit. The phase-coexistence runs contain 27 $MgSiO_3$ formula units and 92 to 219 hydrogen atoms; the homogeneous runs contain 135 to 160 atoms. All compositions are H_2 mole fractions x on the oxide basis (Eq. 2.6).

B.1 Phase-Coexistence Runs

Table B.1 lists all 14 phase-coexistence runs. For runs that kept two phases it gives the composition of the H-poor and H-rich phase from ED Table 1 and, where they differ, from the digitized Figure 2a. For runs that became homogeneous it gives the bulk composition in parentheses. The last column is the published critical temperature at the run pressure minus the run temperature: positive values mean the model places the run below T_C.

In every two-phase run the per-phase atom numbers of ED Table 1 add up to the full simulation cell, to within one atom. Each atom, including those in the two interfaces, is therefore assigned to one of the two phases. For the 2.05 and 9.58 GPa runs the table and the figure agree to within 0.0006. For the 1.65 and 5.14 GPa runs, which lie closest to the critical curve, they differ by up to 0.07.

Table B.1: ***Phase-coexistence runs of ED Table 1 and the published fit.*** *Pressure uncertainties in the last digits are given in parentheses. "same" means that the digitized Figure 2a agrees with the table to within 0.001.*

P (GPa)	T (K)	Phases	ED Table 1 x	Figure 2a x	Published fit x	$T_C - T$ (K)
2.05(4)	3000	two	0.209, 0.931	same	0.231, 0.990	+484
1.65(24)	3500	two	0.458, 0.576	0.419, 0.629	0.595, 0.828	+26
1.89(22)	4000	one	(0.522)	–	one phase	−500
1.79(12)	5000	one	(0.460)	–	one phase	−1489
1.48(7)	6000	one	(0.471)	–	one phase	−2456
1.41(3)	8000	one	(0.565)	–	one phase	−4449
5.14(10)	3000	two	0.587, 0.706	0.514, 0.736	0.398, 0.941	+157
4.71(5)	3500	one	(0.616)	–	one phase	−298
4.06(5)	4000	one	(0.597)	–	one phase	−729
3.95(5)	5000	one	(0.583)	–	one phase	−1717
9.58(10)	2500	two	0.316, 0.976	same	0.346, 0.961	+187
10.42(6)	3000	one	(0.670)	–	one phase	−401
9.29(14)	4000	one	(0.660)	–	one phase	−1282
7.85(28)	6000	one	(0.638)	–	one phase	−3130

B.2 Homogeneous Excess Properties

Table B.2 lists the excess properties reconstructed from ED Table 2 per mole of mixing units. The uncertainties include the shared endmember errors and the pressure-offset systematic described in Section 2.5. The enthalpy entries of different compositions are correlated with coefficients of 0.48 to 0.76, the volumes with 0.48 to 0.78.

Table B.2: ***Excess properties of the homogeneous simulations at 5.06–5.08 GPa and 4,000 K*** *(ED Table 2), with 1σ uncertainties, next to the published fit.*

	H_{ex} (kJ mol^{-1})		S_{ex} (J mol^{-1} K^{-1})		V_{ex} (cm^3 mol^{-1})	
x	simulation	fit	simulation	fit	simulation	fit
0.833	37.4 ± 3.7	43.5	10.6 ± 3.1	11.5	-0.36 ± 0.13	−1.45
0.588	63.1 ± 4.4	58.0	13.9 ± 4.9	15.4	-1.00 ± 0.17	−1.94
0.238	37.7 ± 6.0	24.5	7.6 ± 6.1	6.5	-0.44 ± 0.27	−0.82

C. Recalibration Details

This appendix documents the priors, the sampler and the best-fitting parameters of the recalibration of Section 2.5.

C.1 Priors

All priors are uniform. For class M1, A lies between -300 and 1,500 kJ mol^{-1} and B between -300 and 3,000 kJ mol^{-1}. For the other classes the leading coefficients lie between -300 and 3,000 kJ mol^{-1}, and higher Redlich–Kister terms within $\pm$2,000 kJ mol^{-1}. The inverse temperature scale $1/\tau$ lies between 0 and 10^{-3} K^{-1} and the inverse pressure scale $1/\pi$ between -0.25 and 0.1 GPa^{-1}. For class M2 the subregular entropy coefficients lie between -200 and 800 J mol^{-1} K^{-1} and the volume coefficients between -60 and 30 cm^3 mol^{-1}. Parameter sets for which the homogeneous free energy develops more than one unstable composition interval at any evaluated state are outside every model class and receive zero probability.

C.2 Sampler and Convergence

Each fit starts from the best of eight Nelder–Mead optimizations around the published parameters. Eight adaptive Metropolis chains [11] then run 12,000 steps each. During the first 3,000 steps the proposal covariance and scale adapt; these steps are discarded. Every 30th later step is kept, which gives 2,400 posterior samples per fit. The acceptance rate is about 0.2 in every fit. The largest split-chain $\hat{R}$ over all parameters and derived quantities of all fits is 1.05.

C.3 Best-Fitting Parameters

Table C.1 lists the best-fitting (MAP) parameters of class M1 for each data treatment, next to the published values.

Table C.1: ***Best-fitting parameters of the published model form (class M1).*** *B/A sets the asymmetry and the critical composition; τ and π set the temperature and pressure scales.*

Fit	A (kJ mol^{-1})	B (kJ mol^{-1})	B/A	τ (K)	π (GPa)
Published fit	75	424	5.65	4400	−35
M1, all data	198	306	1.55	4260	−46
M1, Fig. 2a compositions	199	302	1.52	4293	−46
M1, $\sigma_x = 0.15$ near T_C	140	332	2.38	4427	−55
M1, without volumes	234	356	1.52	4260	−37
M1, phases + compositions	226	1378	6.10	3686	−35
M1, phases + homogeneous	233	326	1.40	4488	−66

D. Reproducibility

This appendix lists how every number and figure in this draft is produced. All paths are relative to the repository root. The analysis used Python 3.13 with the package versions recorded in `research/spinodal_onset/outputs/provenance.json`.

D.1 Pipeline

1. `run_study.py` computes the onset grid, the cooling paths, the source-data audit, the homogeneous audit, the stress cases and the native `Atmodeller` solves.
2. `identifiability.py` builds the perturbation of Eq. (2.13) and checks that it leaves every equilibrium quantity unchanged.
3. `digitize_source_figure.py` digitizes Figure 2a of [9] and compares it with our implementation.
4. `calibration.py` runs the recalibration (Section 2.5); about 40 minutes of computation with eight worker processes.
5. `design.py` computes the design screen of Section 4.3.
6. `make_paper_figures.py` writes all figures, and `paper/make_numbers.py` writes the computed numbers and table bodies to `paper/numbers.tex` and `paper/tables`.

Most computed numbers in the text are macros generated from the output files, so rebuilding the draft after a rerun updates the text with the figures. A few values are typed directly, such as the bulk compositions of source runs and rounded ratios; they are listed with their origin in Appendix B or follow from the macros.

D.2 Commands

```
uv sync --locked --all-extras --dev
PYTHONPATH=. uv run python research/spinodal_onset/run_study.py
PYTHONPATH=. uv run python research/spinodal_onset/identifiability.py
PYTHONPATH=. uv run python research/spinodal_onset/digitize_source_figure.py
PYTHONPATH=. uv run python research/spinodal_onset/calibration.py
PYTHONPATH=. uv run python research/spinodal_onset/design.py
PYTHONPATH=. uv run python research/spinodal_onset/make_paper_figures.py
PYTHONPATH=. uv run python research/spinodal_onset/paper/make_numbers.py
cd research/spinodal_onset/paper && tectonic -X compile main.tex
```

D.3 Data Provenance

The ED Tables and Figure 2 of [9] are stored unmodified in `research/spinodal_onset/sources` with their SHA-256 hashes, source URLs and the CC-BY-4.0 attribution. No source value was changed. Where the table and the figure disagree, both are kept and analysed as separate data versions. The thesis layout reuses the KU Leuven title-page logo of the bachelor thesis. Its use should be checked against the current KU Leuven style rules before the draft is submitted or shared beyond this repository.

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